



MolParetoBO

Surrogate-assisted multi-objective molecular optimization –
living technical report

STUDENT

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PROJECT

Project 06 • MolParetoBO

DOCUMENT

Technical report

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1. Goal and pipeline

1 • Mutate

RDKit

2 • Filter

suitability

3 • Score

5 endpoints

4 • Surrogate

RF + fingerprints

5 • Pareto

acquire next

Expensive oracles cannot score every mutant. The surrogate proposes cheap predictions; acquisition chooses which molecules deserve a true oracle call.

2. Design decisions

ITEM	DECISION	RATIONALE
Seed	Ibuprofen	Build the pipeline
Endpoints	5 objectives	QED, SA, logP, logS, Tox
Mutations	Methyl and ring edits	Also applied again to some mutants to make more molecules
Filters	Suitability gates	sanitize, MW, QED
Surrogate	RF + Morgan FP	cheap endpoint prediction
Acquisition	quality + uncertainty + diversity	simplified vs qEHVI
Runtime	Local + Google Colab	interactive widgets

3. Seed, endpoints, mutations

206

SEED MW

0.82

SEED QED

92

MOLECULES SCORED

92

PASSED FILTERS

SMILES: CC(C)Cc1ccc(cc1)C(C)C(=O)O

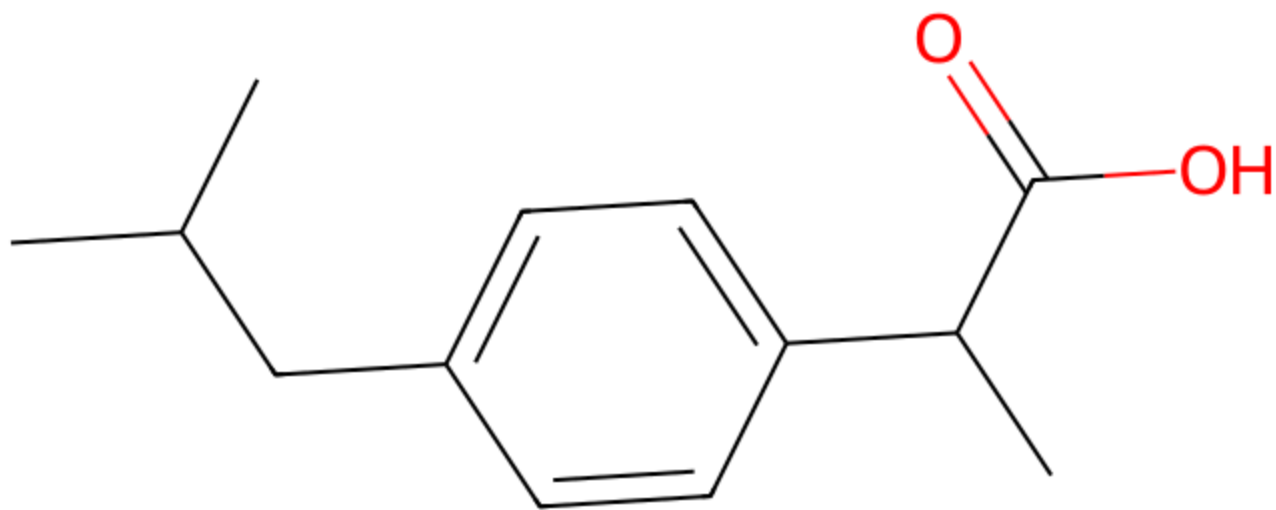


Figure 1. Ibuprofen with atom indices.

#	SYMBOL	MEANING	GOAL	ORACLE
1	QED	Drug-likeness	maximize	RDKit QED
2	SA	Synthetic accessibility	minimize	RDKit SA Score
3	logP	Lipophilicity	prefer ~2.5	Crippen logP
4	logS	Aqueous solubility	maximize	ESOL-style logS
5	Tox	Toxicity proxy	minimize	Transparent heuristic

Suitability after mutation: sanitize · deduplicate · MW 150–450 · heavy atoms ≥ 8 · QED ≥ 0.3 .

FAMILY	GENERATED	KEPT
methyl_add	37	37
methyl_remove	12	12
ring_sub	42	42
seed	1	1

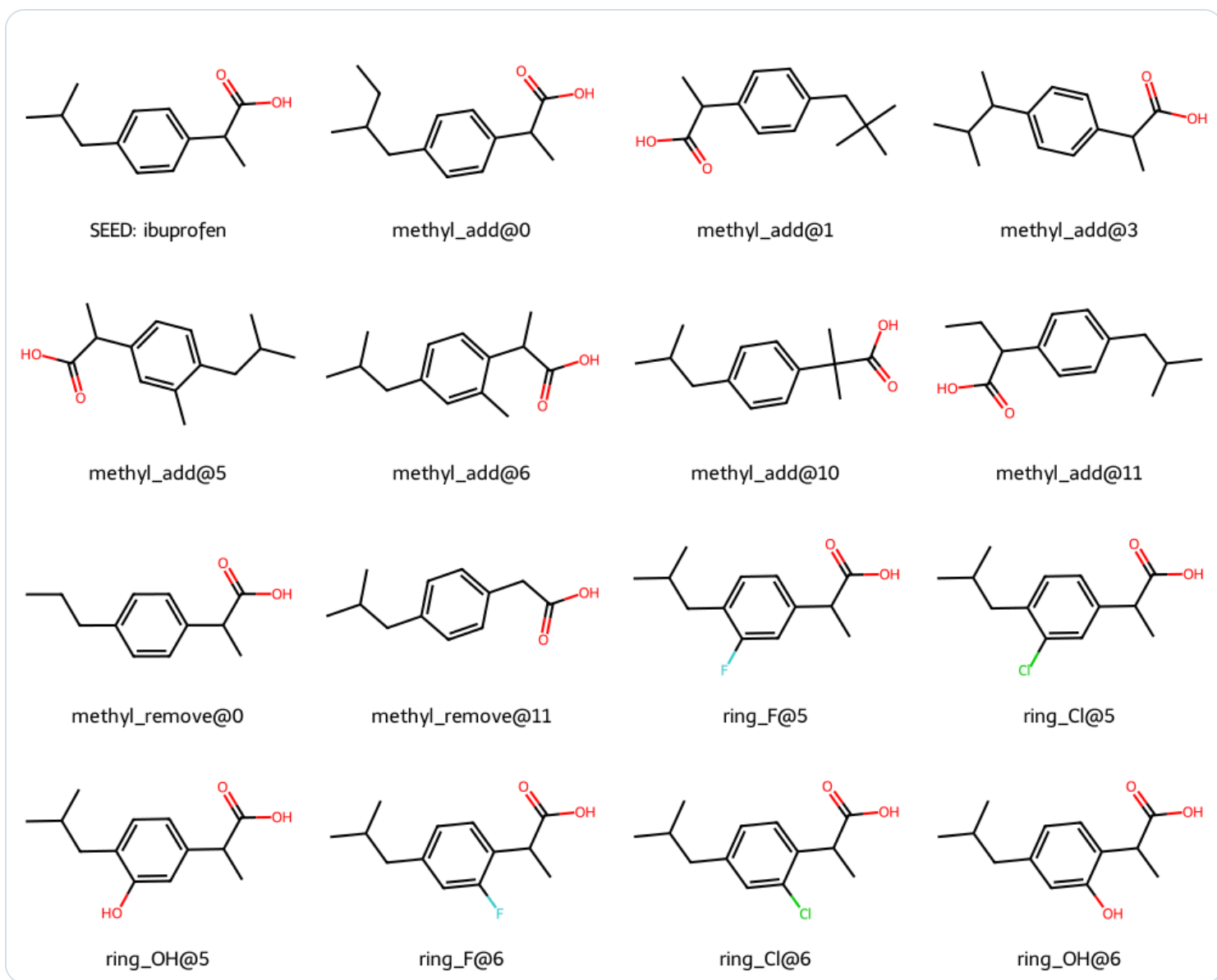


Figure 2. Mutation gallery (seed + kept neighbours).

Scored preview

MUTATION	QED	SA	LOGP	LOGS	TOX	MW	KEPT
SEED	0.822	2.192	3.073	-3.531	0.000	206.285	True
methyl_add@0	0.825	2.602	3.463	-3.798	0.000	220.312	True
methyl_add@1	0.847	2.398	3.463	-3.930	0.000	220.312	True
methyl_add@3	0.841	2.731	3.634	-3.971	0.000	220.312	True
methyl_add@5	0.844	2.458	3.382	-3.812	0.000	220.312	True
methyl_add@6	0.844	2.464	3.382	-3.812	0.000	220.312	True
methyl_add@10	0.845	1.917	3.247	-3.728	0.000	220.312	True
methyl_add@11	0.825	2.269	3.463	-3.798	0.000	220.312	True
methyl_remove@0	0.796	2.124	2.827	-3.289	0.000	192.258	True
methyl_remove@11	0.795	1.559	2.512	-3.091	0.000	192.258	True
ring_F@5	0.852	2.477	3.212	-3.730	0.000	224.275	True
ring_Cl@5	0.871	2.461	3.727	-4.156	0.000	240.730	True

4. Surrogate model

Morgan fingerprints (radius 2, 2048 bits) → multi-output Random Forest for QED, SA, logP, logS, Tox.

ENDPOINT	MAE	R2
qed	0.008	0.791
sa	0.108	0.787
logp	0.155	0.766
logs	0.128	0.759
tox	0.000	1.000

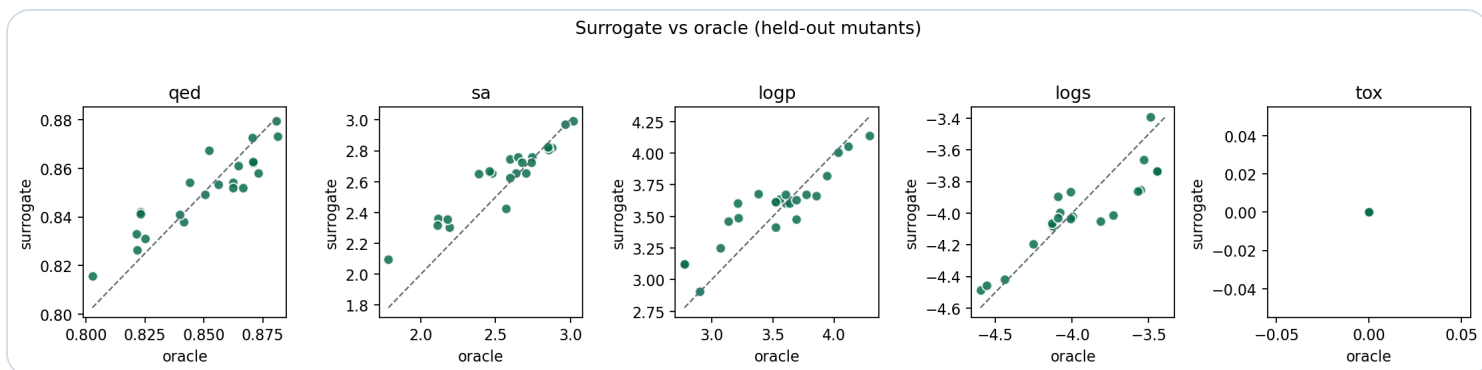


Figure 3. Surrogate vs oracle (held-out parity).

5. Pareto front

Objectives maximized after sign flips: +QED, -SA, -|logP-2.5|, +logS, -Tox. Non-dominated molecules form the approximate front.

20

FRONT SIZE (ALL KEPT)

13

FRONT AFTER ACTIVE
LOOP

5

OBJECTIVES

2D

HV MONITOR

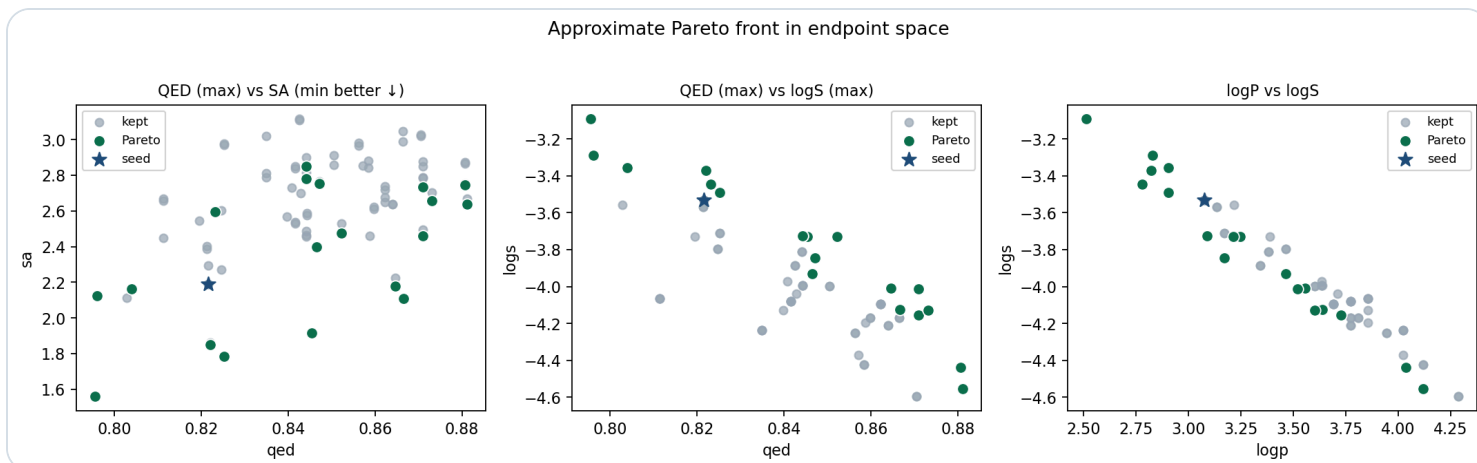


Figure 4. Pareto projections in endpoint space.

Front preview

MUTATION	QED	SA	LOGP	LOGS	TOX	MW
methyl_add@1	0.847	2.398	3.463	-3.930	0.000	220.312
methyl_add@10	0.845	1.917	3.247	-3.728	0.000	220.312
methyl_remove@0	0.796	2.124	2.827	-3.289	0.000	192.258
methyl_remove@11	0.795	1.559	2.512	-3.091	0.000	192.258
ring_F@5	0.852	2.477	3.212	-3.730	0.000	224.275
ring_Cl@5	0.871	2.461	3.727	-4.156	0.000	240.730
ring_OH@6	0.823	2.597	2.779	-3.445	0.000	222.284
gen2 methyl_remove@10	0.804	2.162	2.902	-3.357	0.000	206.285
gen2 methyl_add@1	0.867	2.111	3.637	-4.126	0.000	234.339
gen2 methyl_remove@0	0.825	1.783	2.902	-3.489	0.000	206.285
gen2 ring_F@7	0.873	2.657	3.602	-4.129	0.000	238.302
gen2 ring_Cl@7	0.881	2.637	4.117	-4.555	0.000	254.757

6. Acquisition + iterative loop

Acquisition = $\alpha \cdot \text{quality} + \beta \cdot \text{uncertainty} + \gamma \cdot \text{diversity}$ (defaults 0.5 / 0.3 / 0.2). Each iteration selects a batch, adds oracle labels, retrains the surrogate, and updates the front.

ITER	N_LABELLED	FRONT_SIZE	HV_QED_NEGSA	BATCH_MEAN_ACQ
1	13	9	1.824	0.691
2	17	10	1.830	0.595
3	21	12	1.844	0.662
4	25	12	1.844	0.688
5	29	13	1.844	0.659

Active loop progress

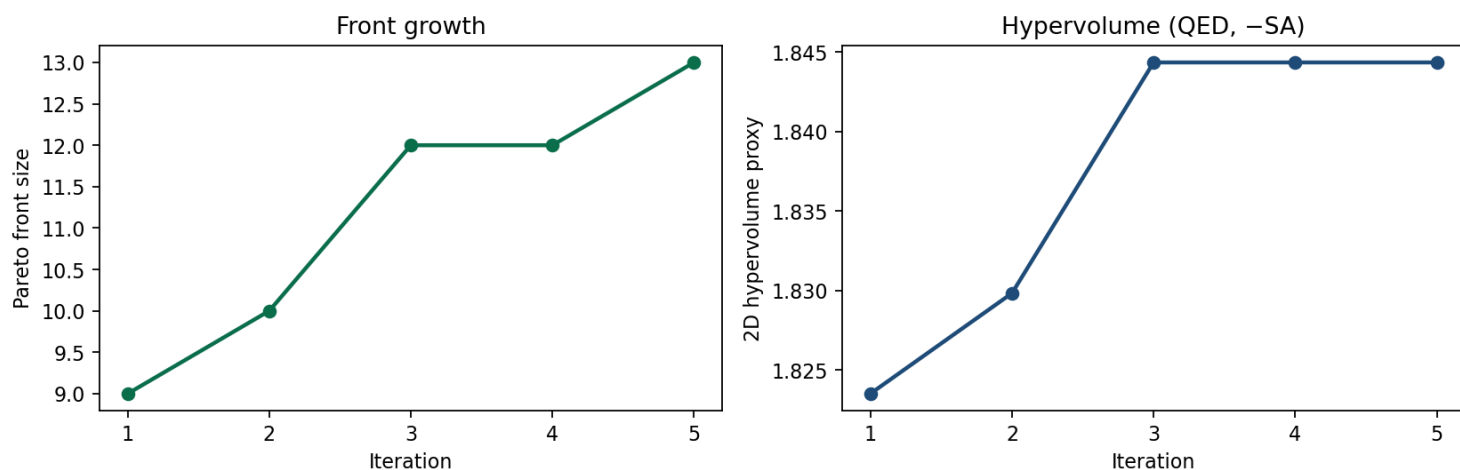


Figure 5. Active-loop progress (front size and 2D hypervolume proxy).

Interactive widgets in the notebook: tune QED filter, run pipeline, query surrogate, and adjust acquisition weights (α , β , γ) live – on local Jupyter or Google Colab.

7. Critical discussion

Scientific question: can a small surrogate-assisted loop expose trade-offs among drug-likeness, SA, lipophilicity, solubility and a toxicity proxy from one public seed? **Yes**, as a transparent proof of concept.

- **Interpretation:** Pareto molecules are undominated compromises, not “the best drug”.
- **AI relevance:** fingerprints + learning surrogate + active multi-objective selection.
- **Assumptions:** computational oracles; pedagogical tox; local mutations; RF variance as uncertainty.
- **Limits:** no wet-lab truth; small neighbourhood; 2D HV proxy; scalar acquisition $\neq$ hypervolume-optimal.
- **SoftMining link:** same industrial pattern (multi-objective design + selective expensive evaluation) at teaching scale.

8. Extensions — GP, NSGA-II, EHVI

Beyond the minimal project: runnable upgrades that point toward research-grade Bayesian optimization.

GP

PCA + UNCERTAINTY

NSGA

CROWDING SUBSET

EHVI

2D MC DEMO

qEHVI

FUTURE (BOTORCH)

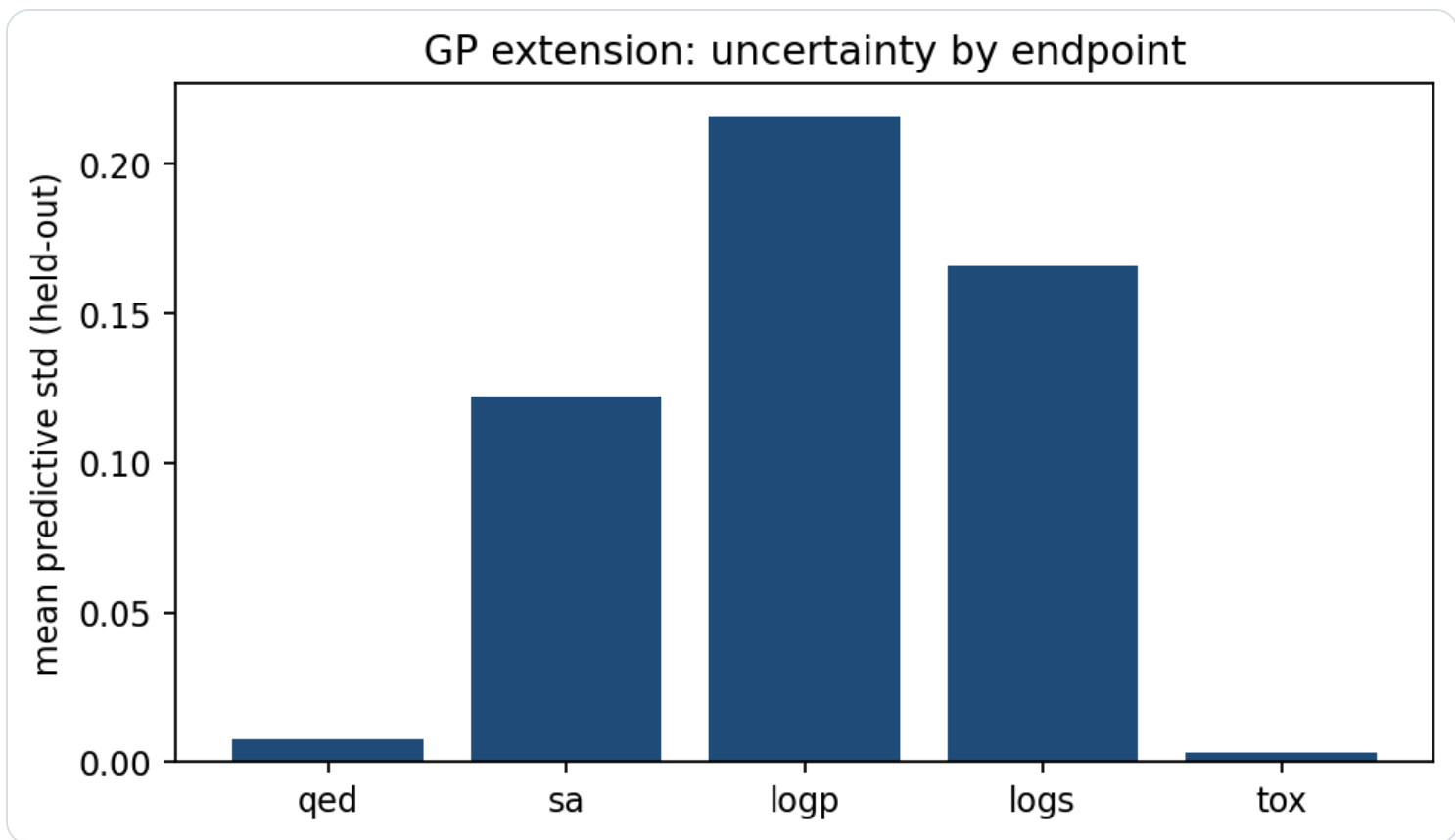


Figure 6. GP extension: mean predictive uncertainty by endpoint.

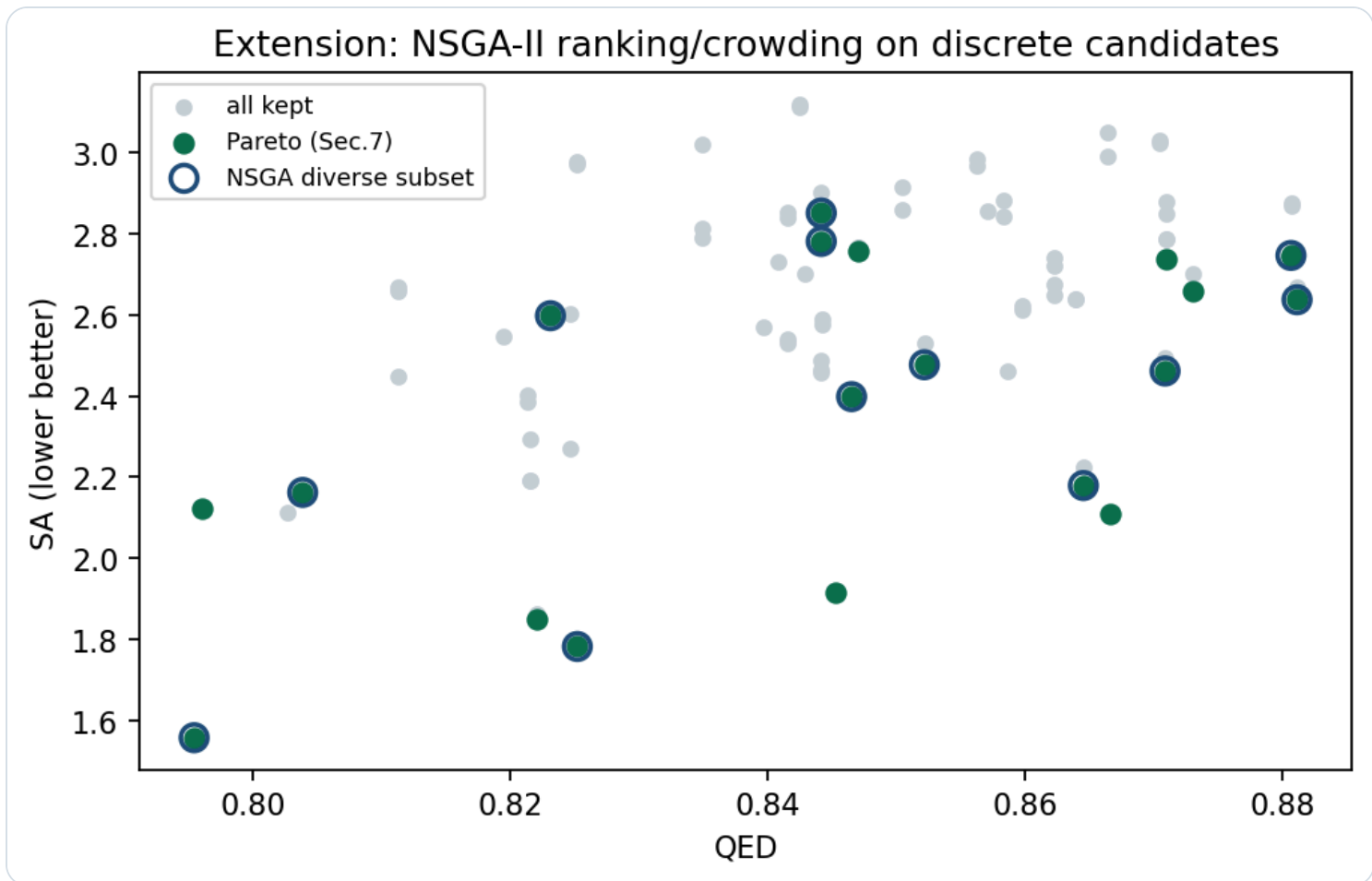


Figure 7. NSGA-II non-dominated sorting + crowding on the mutant pool.

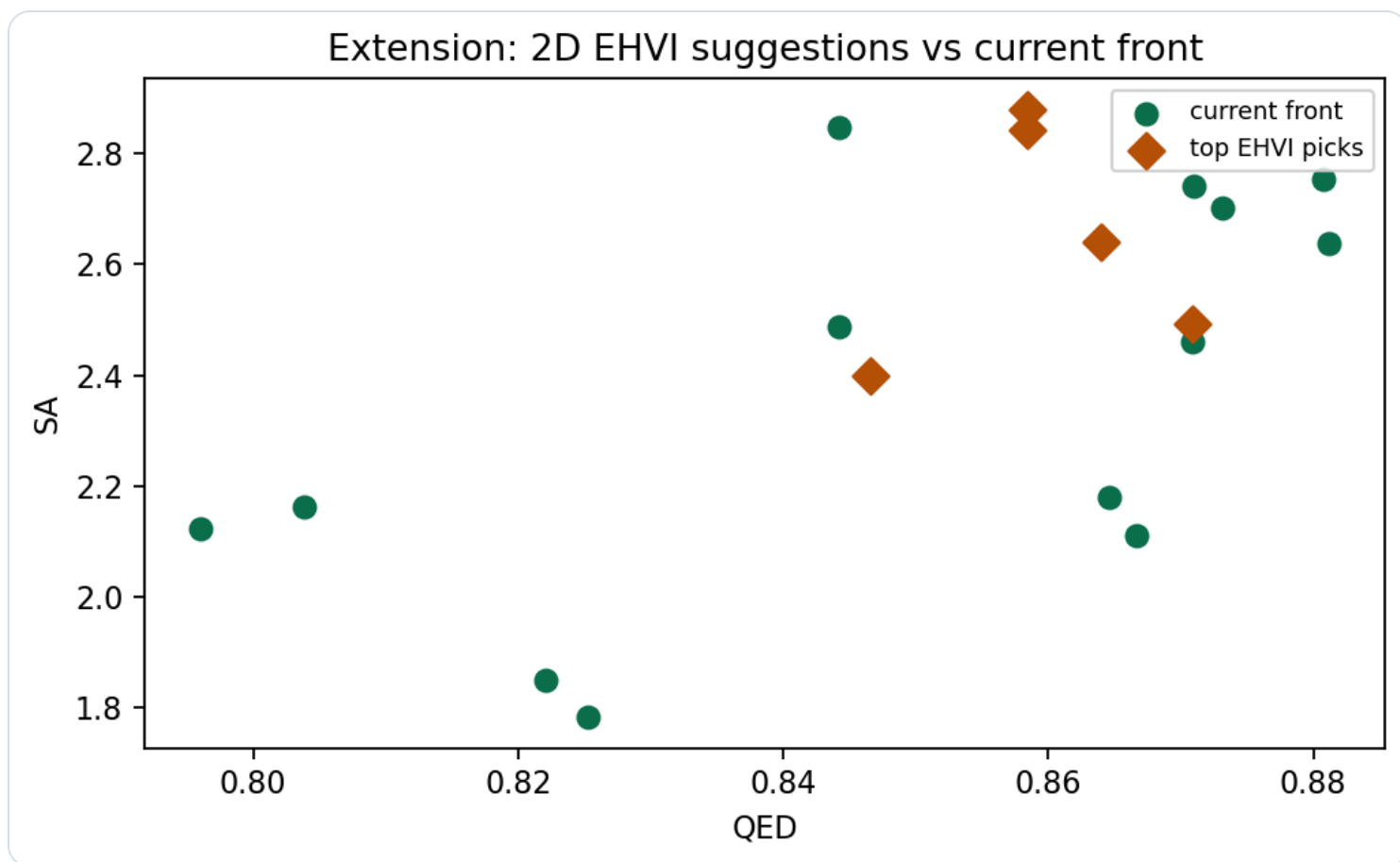


Figure 8. 2D EHVI suggestions relative to the current front.

Presentation: use `MolParetoB0_Slides.pdf` / `MolParetoB0_Slides.pptx` (about 12 slides, 10-15 min). Regenerate with `python build_slides.py`.

9. Interactive widgets & tools

The notebook includes interactive **ipywidgets** so you can verify runnability without editing code.

Interactive Control Panel (Section 8c)

BUTTON	WHAT IS CHECKED
1 • Run filter check	Mutations + suitability filters
2 • Inspect molecule	Structure drawing + oracle endpoints
3 • Query surrogate	Fingerprint → RF vs oracle
4 • Select batch	Acquisition ranking ($\alpha/\beta/\gamma$)
5 • Pareto summary	Front size + HV proxy

Earlier widgets also exist in Sections 5b (pipeline), 6 (surrogate), and 8b (acquisition weights).

Tools used (summary)

TOOL	ROLE
RDKit	SMILES, mutations, QED/logP/MW, fingerprints, drawings; SA Score via Contrib
scikit-learn	Random Forest surrogate; PCA + Gaussian Process (extension); metrics
pymoo	NSGA-II non-dominated sorting / crowding (extension)
NumPy / pandas / matplotlib	Numerics, tables, figures; custom 2D EHVI demo
ipywidgets + Jupyter/Colab	Interactive checks
python-pptx + Chrome PDF	Slides and styled report export

Full version table and oracle method details: **Appendix B** in the notebook.

Companion: MolParetoBO_Report.ipynb · slides: MolParetoBO_Slides.pdf · 05 August 2026